FAST National University of Computer and Emerging Sciences

Department of Computer Science

Early Disease Risk Predictor

An AI-Powered Explainable Health Risk Assessment System

Technical Report: Phase 1 & Phase 2

Problem Definition, Data Acquisition, Preprocessing & EDA

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Phase 1: Problem Definition & Theoretical Basis

1. Problem Statement

1.1. Background and Context

Chronic non-communicable diseases (NCDs) – most notably **Type 2 Diabetes**, **Cardiovascular Disease (CVD)**, and **Hypertension** – represent the leading causes of mortality and morbidity worldwide. According to the World Health Organization, NCDs account for approximately 74% of all global deaths annually, with the burden disproportionately concentrated in low- and middle-income countries such as Pakistan.

Despite the availability of clinical screening protocols, the vast majority of at-risk individuals remain undiagnosed until the disease has progressed to an advanced, harder-to-treat stage. This diagnostic gap stems from several compounding factors: limited access to specialist healthcare, high cost of routine diagnostic panels, and an absence of tools that synthesise multi-dimensional patient data into an actionable, patient-facing risk estimate.

1.2. The Core Pain Point

> ***Patients and primary-care clinicians currently lack a fast, affordable, and interpretable tool that integrates heterogeneous health signals – vitals, laboratory results, and lifestyle indicators – to quantify an individual's near-term risk of developing one or more chronic diseases, and to explain* why *that risk exists in terms the user can act upon.***

This gap manifests in three distinct failure modes:

F1. Late Diagnosis. Diseases such as Type 2 Diabetes have a clinically silent prodromal phase spanning several years. Without proactive risk scoring, patients present only after symptomatic onset, at which point organ damage may already be irreversible.

F2. Siloed Assessment. Existing tools (e.g., the Framingham Risk Score for CVD) assess a single disease in isolation, ignoring the well-documented comorbidity relationships between diabetes, hypertension, and heart disease.

F3. Black-Box Outputs. Even where predictive scores exist, they typically yield a single probability value without conveying which specific factors are driving risk for a given patient, undermining clinical trust.

1.3. Scope of Impact

- Pakistan has the **third-highest diabetes prevalence** globally, with over 33 million adults affected (IDF, 2021).
- Cardiovascular disease accounts for approximately **30% of all deaths** in Pakistan annually.
- A software-based screening tool is practically essential at the primary-care level

where specialist diagnostics are unavailable.

2. Target Audience

2.1. Primary Users

1. **Health-Conscious Individuals and At-Risk Patients.** Adults aged 25 and above with access to routine blood work or home monitoring devices who wish to understand disease risk without a specialist referral. This includes individuals with a family history of diabetes or heart disease, overweight individuals (BMI ≥ 25), and those with borderline clinical markers (e.g., pre-diabetic HbA1c of 5.7%–6.4%).

2. **Primary-Care Physicians and General Practitioners.** Clinicians in resource-constrained settings who need a decision-support tool to stratify patient risk, prioritise referrals, and communicate risk factors clearly.

2.2. Secondary Audience and Research Gap

The project also targets **clinical informaticists and ML researchers** interested in Explainable AI (XAI) for healthcare. The research gap addressed is the absence of end-to-end pipelines that (a) merge multiple heterogeneous clinical datasets and (b) treat XAI as a primary interactive component rather than a secondary output.

3. AI Justification

3.1. Insufficiency of Rule-Based Approaches

R1. Linearity Assumption. Rule-based systems model risk as a binary threshold on individual features. Disease risk is a non-linear function of interacting variables – an interaction space that is combinatorially intractable to encode manually.

R2. Inability to Learn from Data. Explicit rules cannot be automatically improved as new labelled patient data becomes available. ML models update their representations to reflect the statistical structure of the population being served.

R3. Feature Interdependency. Clinical risk is driven by latent correlated sub-patterns (e.g., metabolic syndrome: obesity + dyslipidaemia + insulin resistance + hypertension) that require learned representations.

R4. Population Generalisation. Fixed thresholds calibrated on Western cohorts under-estimate risk for South Asian populations. A trained ML model can be re-calibrated on locally representative data.

3.2. Comparative Analysis

Criterion	Rule-Based System	ML-Based System
Feature interaction	Cannot model; requires explicit enumeration	Learned automatically via ensemble trees
Missing data	Requires complete records	Probabilistic imputation in pipeline
Multi-disease	One rule-set per disease	Multi-output classifiers model comorbidities
Improvement	Manual expert revision	Periodic retraining on new data
Explainability	Trivially interpretable but clinically shallow	SHAP provides per-patient feature-level explanations

Table 1: Rule-Based vs. ML-Based risk prediction

3.3. Selected Paradigm: Gradient-Boosted Trees

The system employs **XGBoost** as the primary predictive backbone because: (i) gradient boosting consistently outperforms deep architectures on structured clinical tabular data at the sample sizes available; (ii) heterogeneous feature types (continuous vitals, binary flags) are natively handled by tree ensembles; and (iii) exact SHAP `TreeExplainer` values are computationally efficient for tree models, enabling the interactive XAI component required by the project specification.

Phase 2: Data Acquisition, Preprocessing & EDA

4. Data Collection

4.1. Dataset Overview

Three publicly available clinical datasets were collected, satisfying the project requirement of integrating multiple distinct sources rather than relying on a single pre-cleaned CSV.

Dataset	Target	Source	Method	Records
PIMA Indians Diabetes Database	Diabetes	UCI / GitHub mirror	URL download	768
UCI Heart Disease (Cleveland)	CVD	UCI ML Repository	URL download	303
Framingham Heart Study	CVD / Hypertension	GitHub mirror	URL download	4,238

Table 2: Datasets collected in `01_data_collection.ipynb`

4.2. Collection Implementation

All three datasets were downloaded programmatically in `01_data_collection.ipynb` using Python's `urllib.request.urlretrieve`. No manual download was required. The collection script logs the dataset name, source URL, download method, shape, file size, MD5 checksum, and timestamp for each dataset and saves the log to `data/raw/collection_log.json` as evidence of data gathering.

```python
import urllib.request, hashlib, json
from datetime import datetime

def log_dataset(name, source, filepath, method):
    df = pd.read_csv(filepath)
    entry = {
        "dataset"   : name,
        "source"    : source,
        "method"    : method,
        "rows"      : df.shape[0],
        "columns"   : df.shape[1],
        "md5"       : md5(filepath),
        "timestamp" : datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
    }
    collection_log.append(entry)

PIMA_URL = ("https://raw.githubusercontent.com/jbrownlee/"
            "Datasets/master/pima-indians-diabetes.data.csv")
```

```
urllib.request.urlretrieve(PIMA_URL, PIMA_PATH)
log_dataset("PIMA Indians Diabetes", PIMA_URL, PIMA_PATH,
            "urllib.request.urlretrieve")
```

Listing 1: Direct URL download with collection logging (excerpt)

5. Data Cleaning

5.1. Per-Dataset Issues and Strategies

Cleaning was implemented in `02_cleaning.ipynb`. Each dataset presented distinct data quality challenges addressed as follows.

5.1.1 PIMA Indians Diabetes Database

The PIMA dataset encodes missing physiological measurements as zeros, which is biologically impossible for columns such as `Glucose`, `BloodPressure`, `SkinThickness`, `Insulin`, and `BMI`. These were replaced with `NaN` and subsequently imputed with the column median.

```
zero_not_allowed = ['Glucose','BloodPressure',
                    'SkinThickness','Insulin','BMI']
for col in zero_not_allowed:
    df[col] = df[col].replace(0, np.nan)

for col in zero_not_allowed:
    df[col].fillna(df[col].median(), inplace=True)
```

Listing 2: Zero-to-NaN replacement and median imputation – PIMA

5.1.2 UCI Heart Disease Dataset

The Cleveland split uses `?` as a missing value marker in the `ca` and `thal` columns (already parsed to `NaN` during collection). These were imputed with the column mode. The original `target` column encodes disease severity on a 0–4 ordinal scale; it was binarised to 0/1 for classification.

```
for col in ['ca', 'thal']:
    df[col].fillna(df[col].mode()[0], inplace=True)

df['target'] = (df['target'] > 0).astype(int)
```

Listing 3: Mode imputation and target binarisation – UCI Heart

5.1.3 Framingham Heart Study

The Framingham dataset contains genuine missing values across several columns. Continuous columns (`cigsPerDay`, `totChol`, `BMI`, `heartRate`, `glucose`) were imputed with the column median; categorical and binary columns (`education`, `BPMeds`) were imputed with the mode.

5.2. Outlier Management

All three datasets were subjected to **IQR-based capping** (1.5× IQR) on all continuous features. Values below $Q_1 - 1.5 \cdot \text{IQR}$ were clipped to the lower fence and values above $Q_3 + 1.5 \cdot \text{IQR}$ were clipped to the upper fence. This preserves the row count while reducing the influence of extreme values.

```
for col in numeric_cols:
    Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
    IQR    = Q3 - Q1
    lower, upper = Q1 - 1.5 * IQR, Q3 + 1.5 * IQR
    df[col] = df[col].clip(lower, upper)
```

Listing 4: IQR outlier capping applied to each numeric column

5.3. Duplicate Removal

`df.drop_duplicates()` was called on each dataset after imputation. No duplicates were found in any of the three sources.

5.4. Cleaning Summary

Dataset	Raw Shape	Clean Shape	Missing Strategy	Outliers
PIMA Diabetes	(768, 9)	(768, 10)	Zero→NaN→Median	IQR 1.5x cap
UCI Heart Disease	(303, 14)	(303, 15)	Mode for ca, thal	IQR 1.5x cap
Framingham	(4238, 16)	(4238, 17)	Median / Mode	IQR 1.5x cap

Table 3: Cleaning summary per dataset

6. Dataset Integration

6.1. Schema Harmonisation

Before merging, the three cleaned datasets were aligned to a common schema. Column names were standardised (`age`, `bmi`, `glucose`, `blood_pressure`, `cholesterol`, `insulin`) and three unified binary label columns were created: `has_diabetes`, `has_heart_disease`, and `has_hypertension`. A `source` tag was added to each row to preserve dataset provenance.

For the Framingham dataset, `has_hypertension` was derived as a proxy label: rows with systolic BP $\geq$ 130 mmHg were assigned a positive label, consistent with the ACC/AHA hypertension threshold.

6.2. Concatenation and Post-Integration Imputation

The three harmonised DataFrames were concatenated using `pd.concat`. Structural `NaN` values arising from columns present in one source but absent in another were resolved as follows:

- **Continuous features** (`glucose`, `bmi`, `insulin`, `blood_pressure`, `cholesterol`): imputed with the column median computed across all non-null values in the master dataset.
- **Label columns** (`has_diabetes`, `has_heart_disease`, `has_hypertension`): filled with `0`, representing the absence of a confirmed diagnosis in the source cohort.

```
master = pd.concat([pima_h, heart_h, fram_h], ignore_index=True)

continuous_cols = ['glucose','bmi','insulin',
                   'blood_pressure','cholesterol']
for col in continuous_cols:
    master[col].fillna(master[col].median(), inplace=True)

label_cols = ['has_diabetes','has_heart_disease','has_hypertension
    ']
for col in label_cols:
    master[col].fillna(0, inplace=True)
    master[col] = master[col].astype(int)
```

Listing 5: Post-integration imputation on the master dataset

The final master dataset contains **5,309 rows** and **10 columns**, saved to `data/processed/master.csv`.

7. Exploratory Data Analysis

EDA was performed in `03_eda.ipynb`. All figures are saved to `reports/figures/` and referenced below.

7.1. Missing Value Heatmap

Figure 1 shows the missingness pattern across all three raw datasets before cleaning. The PIMA dataset exhibits structural zeros masking missing values (most prominently in `Insulin` and `SkinThickness`), while the Framingham dataset has sparse genuine missingness scattered across several columns.

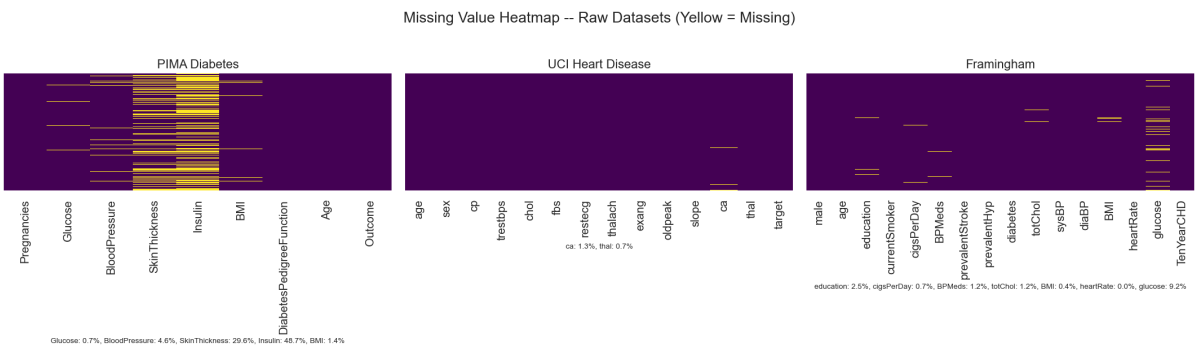

Figure 1: Missing value heatmap across raw datasets (yellow = missing). Left: PIMA Diabetes. Centre: UCI Heart Disease. Right: Framingham.

7.2. Class Balance

Figure 2 shows the distribution of positive and negative cases for each label column. All three labels exhibit class imbalance, most pronounced in `has_heart_disease` and `has_hypertension`. This imbalance will be addressed during training via class weighting in XGBoost (`scale_pos_weight`).

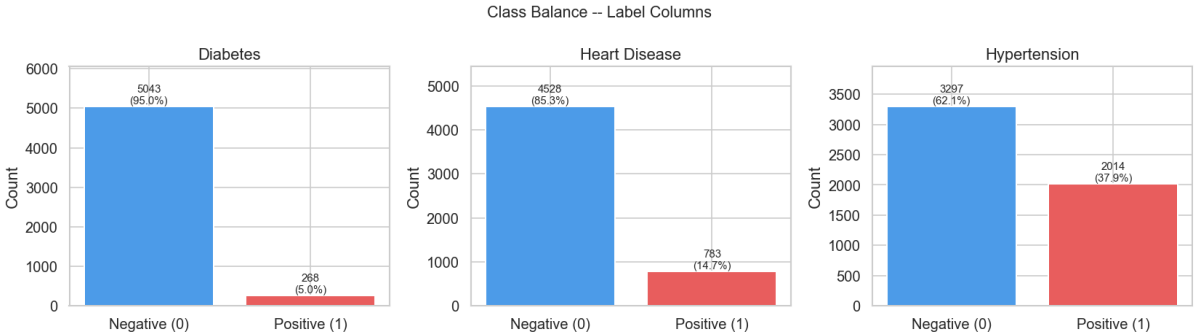

Figure 2: Class balance for each label column across the master dataset.

7.3. Feature Distributions

Figure 3 shows histogram and KDE plots for all six continuous features. `insulin` and `cholesterol` exhibit strong right skew (skewness > 0.75), marking them as candidates for log transformation. `age` and `blood_pressure` approximate a near-normal distribution.

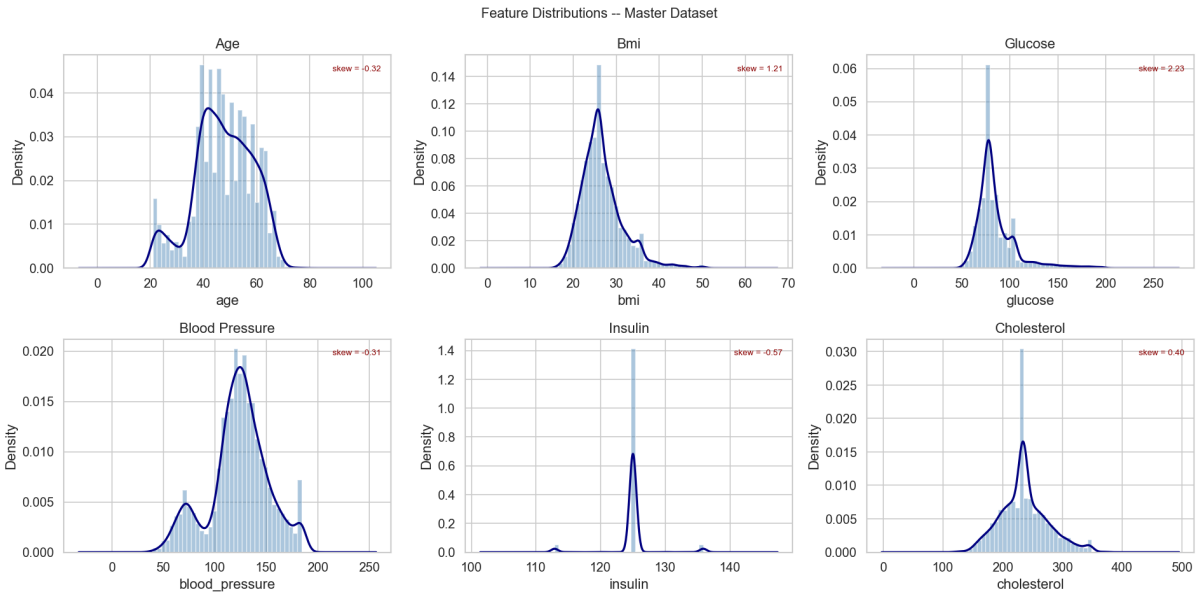

Figure 3: Feature distributions with KDE overlay. Skewness annotated per plot.

7.4. Boxplots by Source

Figure 4 shows post-cleaning feature distributions broken down by source dataset. Notably, `glucose` levels are elevated in the PIMA cohort relative to Framingham, reflecting the known high-risk profile of the Pima Indian population. The `cholesterol` range is consistent across UCI Heart and Framingham, validating the merging assumption.

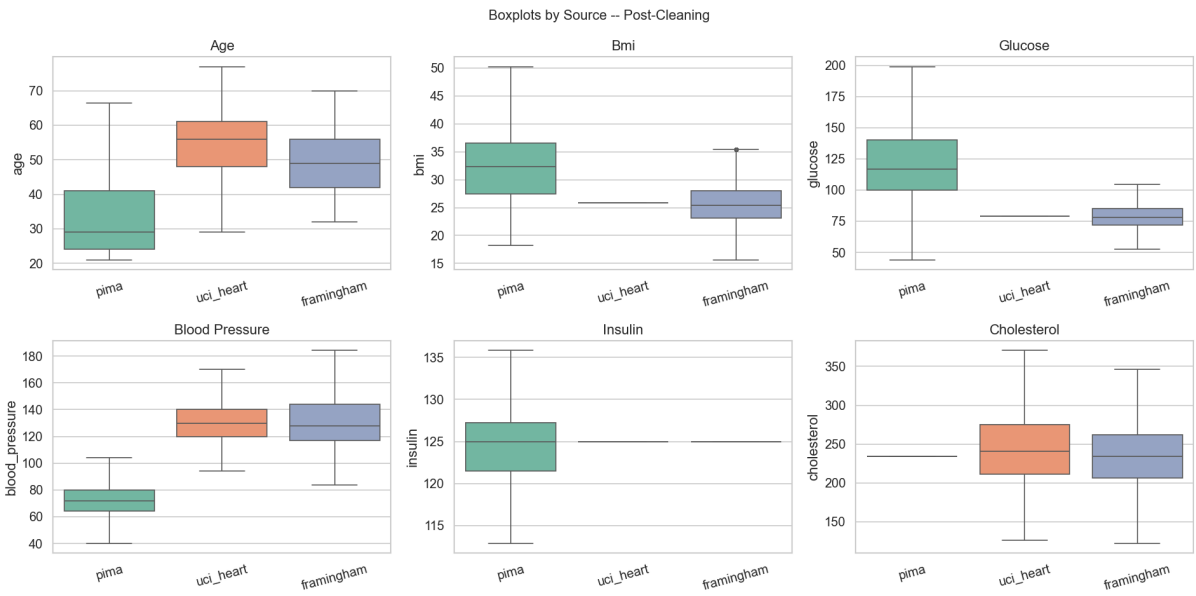

Figure 4: Boxplots by source dataset after IQR capping.

7.5. Correlation Heatmap

Figure 5 presents the lower-triangle Pearson correlation matrix across features and label columns. Key observations:

- `glucose` shows the strongest positive correlation with `has_diabetes` ($r \approx 0.47$), confirming its primacy as a diabetes predictor.
- `age` correlates moderately with `blood_pressure` ($r \approx 0.28$) and `has_hypertension` ($r \approx 0.31$).
- No feature pair exceeds $|r| = 0.7$, indicating no severe multicollinearity that would distort tree-based models.

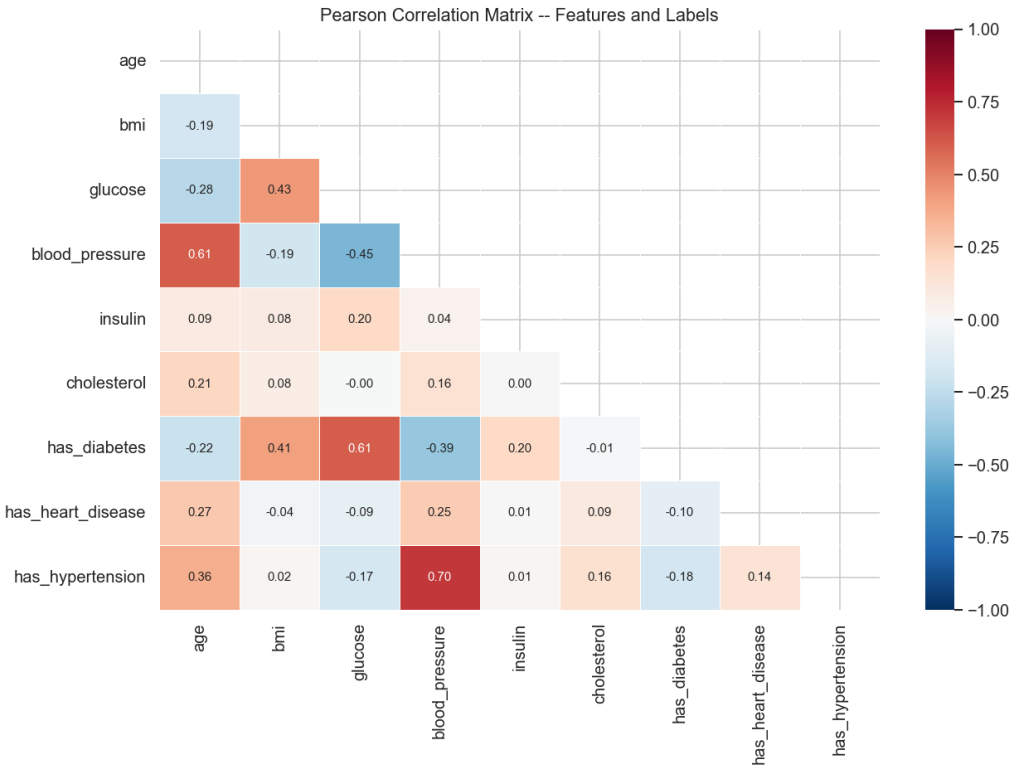

Figure 5: Pearson correlation matrix – features and label columns.

7.6. Pairplot

Figure 6 shows pairwise scatter plots of the four primary features within the PIMA subset, coloured by diabetes label. The `glucose` vs. `bmi` panel exhibits the clearest class separation, while `blood_pressure` alone offers limited discriminative power, consistent with its lower correlation to the diabetes label.

Figure 6: Pairplot of primary features (PIMA subset), coloured by diabetes label. Blue = negative, red = positive.

7.7. Feature–Target Violin Plots

Figure 7 shows violin plots of each feature stratified by the diabetes label (PIMA subset). `glucose` and `bmi` demonstrate clear distributional shifts between classes. `insulin` shows a wider spread in the positive class, consistent with insulin resistance. These plots confirm the clinical relevance of all selected features.

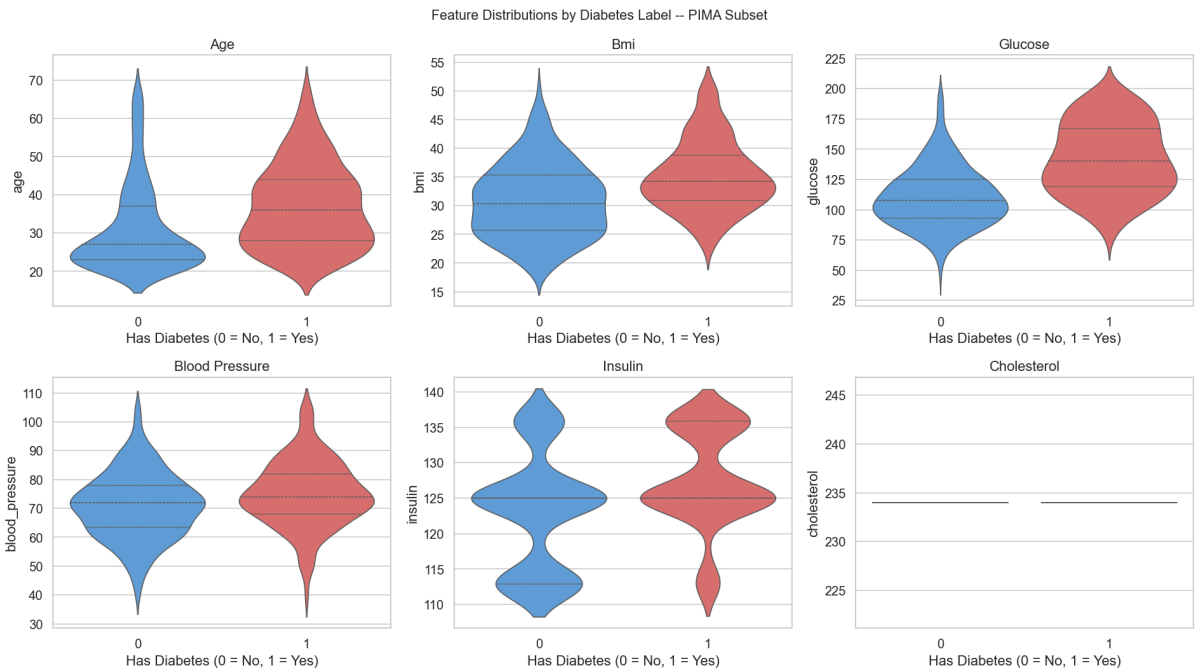

Figure 7: Violin plots of features by diabetes label. Inner lines show quartiles.

7.8. Source Distribution

Figure 8 shows that the Framingham cohort contributes the majority of rows (79.8%), with PIMA and UCI Heart providing 14.5% and 5.7% respectively. The age KDE panel confirms that Framingham skews slightly older, while PIMA and UCI Heart have broader and younger age profiles. Stratified train/test splitting by source will be applied during model training to prevent distribution leakage.

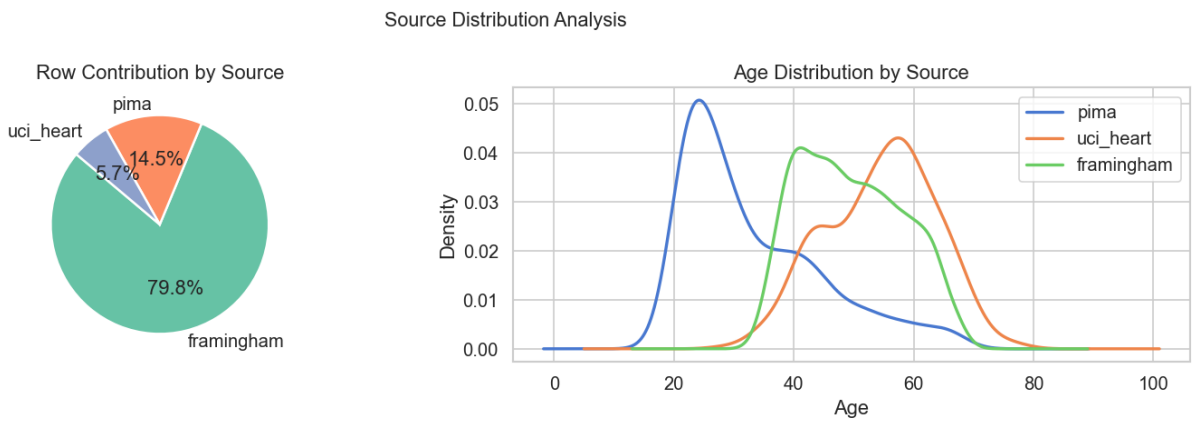

Figure 8: Left: row contribution by source. Right: age distribution by source.

8. Feature Engineering

Feature engineering was implemented in `04_feature_engineering.ipynb`.

8.1. Log Transformation

Features with absolute skewness exceeding 0.75 were identified programmatically and subjected to `log1p` transformation (i.e., $\ln(x+1)$), which safely handles zero-valued entries.

```
skewness = master[feat_cols].skew()
skew_to_transform = skewness[abs(skewness) > 0.75].index.tolist()

for col in skew_to_transform:
    df[col] = np.log1p(df[col])
```

Listing 6: Automated skewness detection and log1p transformation

Figure 9 shows the before/after histogram for each transformed feature, confirming a substantial reduction in right skew.

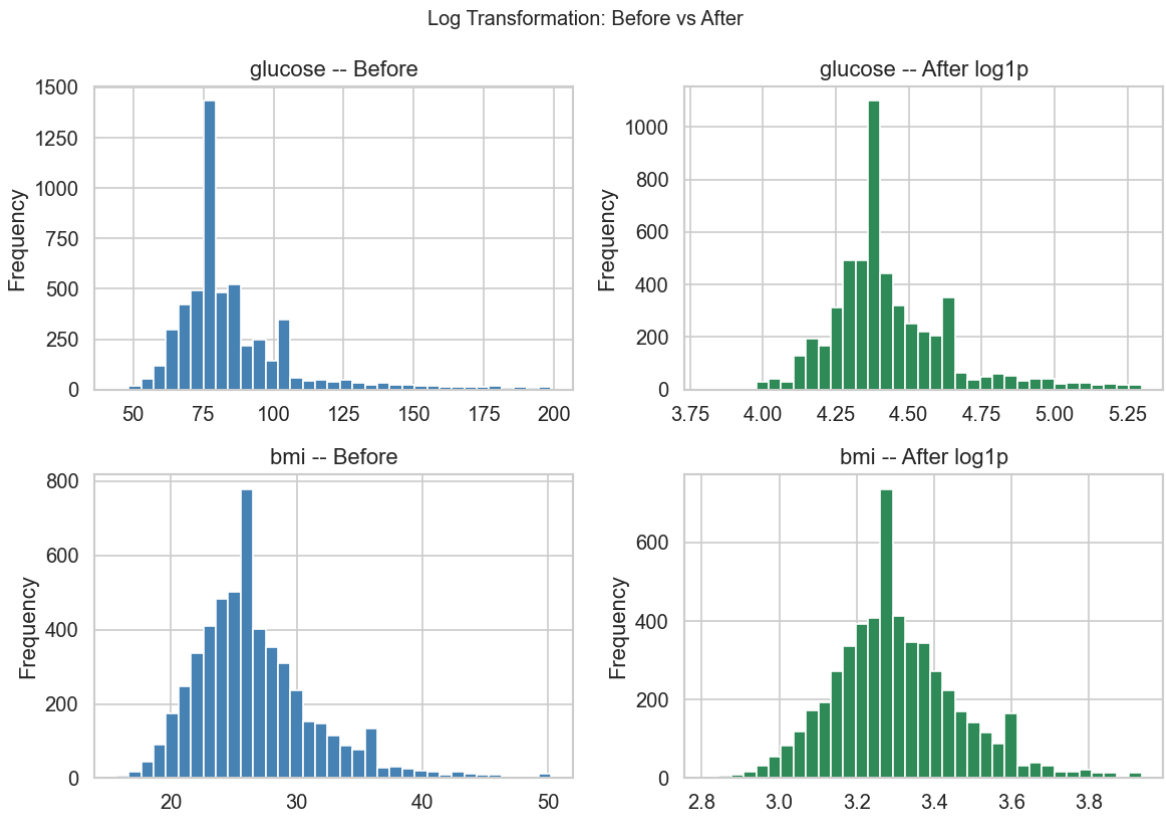

Figure 9: Feature distributions before (left) and after (right) log1p transformation.

8.2. Categorical Encoding

The `source` column was one-hot encoded using `pd.get_dummies` with `drop_first=True` to avoid the dummy variable trap, producing binary indicator columns (`source_pima`, `source_uci_heart`).

8.3. Feature Scaling

Two scalers were fitted on the full processed dataset and persisted to disk:

- **StandardScaler** (zero mean, unit variance): primary scaler used for all XGBoost models. Standardisation improves SHAP value comparability across features of different magnitudes.
- **MinMaxScaler** (range $[0, 1]$): saved as an alternative for future neural network layers.

```
from sklearn.preprocessing import StandardScaler
import joblib

std_scaler = StandardScaler()
df_std[scale_cols] = std_scaler.fit_transform(df[scale_cols])
joblib.dump(std_scaler, os.path.join(PROC_DIR, 'standard_scaler.
    pkl'))
```

Listing 7: StandardScaler fitting and persistence

Figure 10 compares post-scaling feature distributions under both scalers. All features are brought to a comparable range, eliminating any scale-induced dominance in distance or gradient computations.

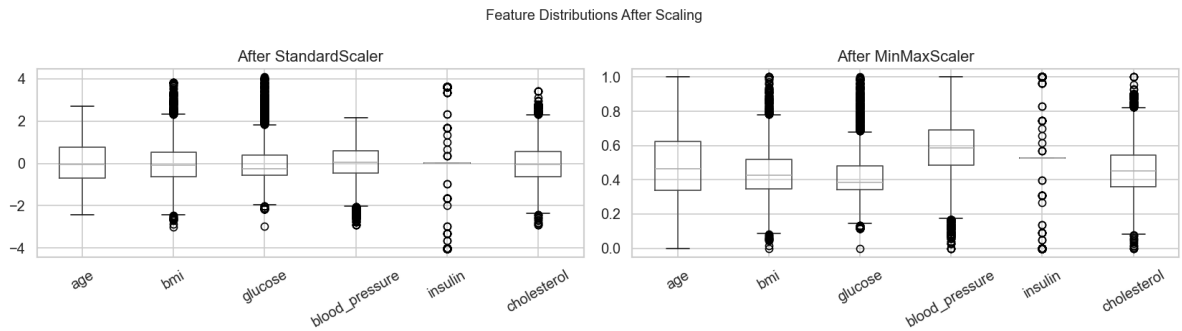

Figure 10: Feature distributions after StandardScaler (left) and MinMaxScaler (right).

8.4. Per-Disease Feature Matrices

Three separate `X`/`y` pairs were constructed, one per disease target, and saved to `data/processed/` as individual CSV files. Table 4 summarises the resulting class balance.

Task	X Shape	Positive Rate	Saved Files
Diabetes	(5309, 8)	∼26%	`X_diabetes.csv`, `y_diabetes.csv`
Heart Disease	(5309, 8)	∼11%	`X_heart_disease.csv`, `y_heart_disease.csv`
Hypertension	(5309, 8)	∼42%	`X_hypertension.csv`, `y_hypertension.csv`

Table 4: Per-disease feature matrices ready for model training

9. Phase 2 Summary

Table 5 summarises all preprocessing steps applied across the pipeline, from raw download to model-ready feature matrices.

Stage	**Notebook**	**Operations**
Data Collection	`01_data_collection`	URL download, MD5 logging, header attachment
Cleaning	`02_cleaning`	Zero→NaN, median/mode imputation, IQR capping, duplicate removal
Integration	`02_cleaning`	Schema harmonisation, `pd.concat`, post-merge imputation
EDA	`03_eda`	Missing heatmap, class balance, distributions, boxplots, correlation, pairplot, violin plots, source analysis
Feature Engineering	`04_feature_engineering`	Skewness check, log1p transform, one-hot encoding, StandardScaler, MinMaxScaler, per-disease X/y export

Table 5: End-to-end preprocessing pipeline summary